

Supplementary Information to:

**Bayesian inference of ancestral host-parasite
interactions under a phylogenetic model of host
repertoire evolution**

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References for host use records for Nymphalini.

Figure S1: Empirical dataset used to validate the model.

Figure S2: Modules of Nymphalini-angiosperms interactions.

Figure S3: Time-calibrated phylogeny of Nymphalini.

Figure S4: Distribution of Bayes factors in the simulation study.

Figure S5: Ancestral state reconstruction of Nymphalini host repertoire.

Figure S6: Marginal posterior densities of parameters in the empirical study.

References used to compile host records for Nymphalini:

- Braby MF (2000) Butterflies of Australia: Their identification, biology and distribution. Vol 2. CSIRO Publishing, Collingwood.
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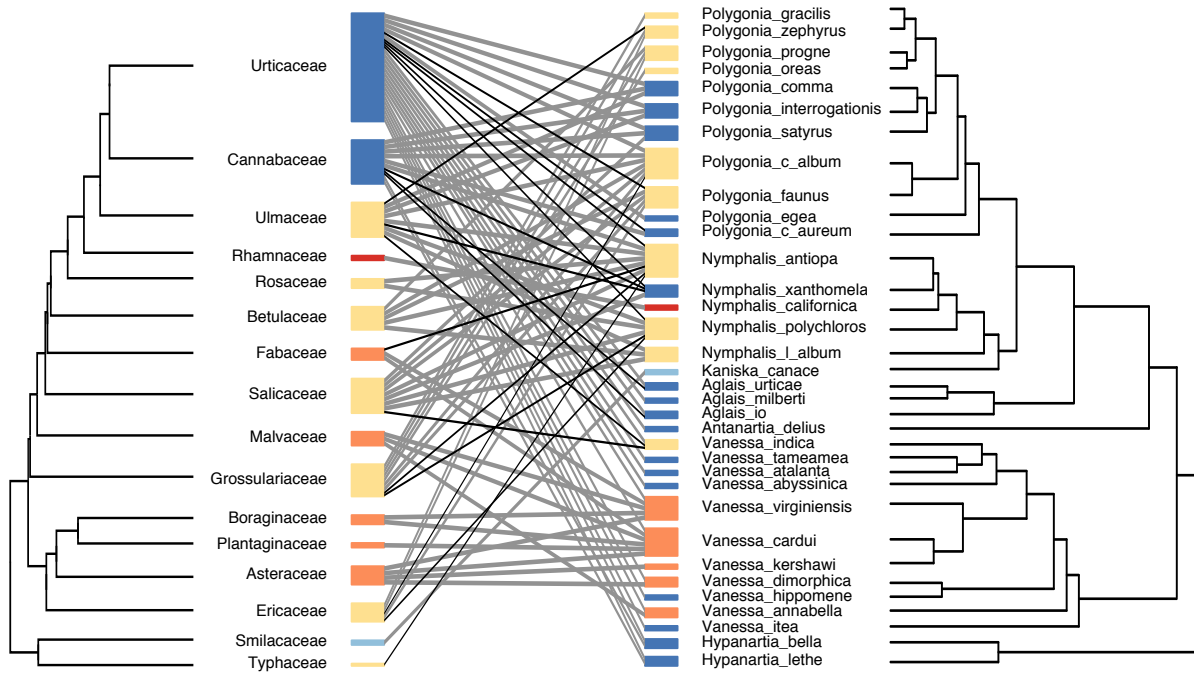


Figure 1: Empirical dataset used to validate the model. Interactions between Nymphalini butterflies (right) and angiosperm families (left) with taxa ordered by phylogenetic relationship. Thickness of boxes is proportional to the number of interactions. Box colors show the five modules found in the dataset (consistent with figs. 7 and S2). Grey lines represent interactions with actual hosts and black lines represent interactions with potential hosts.

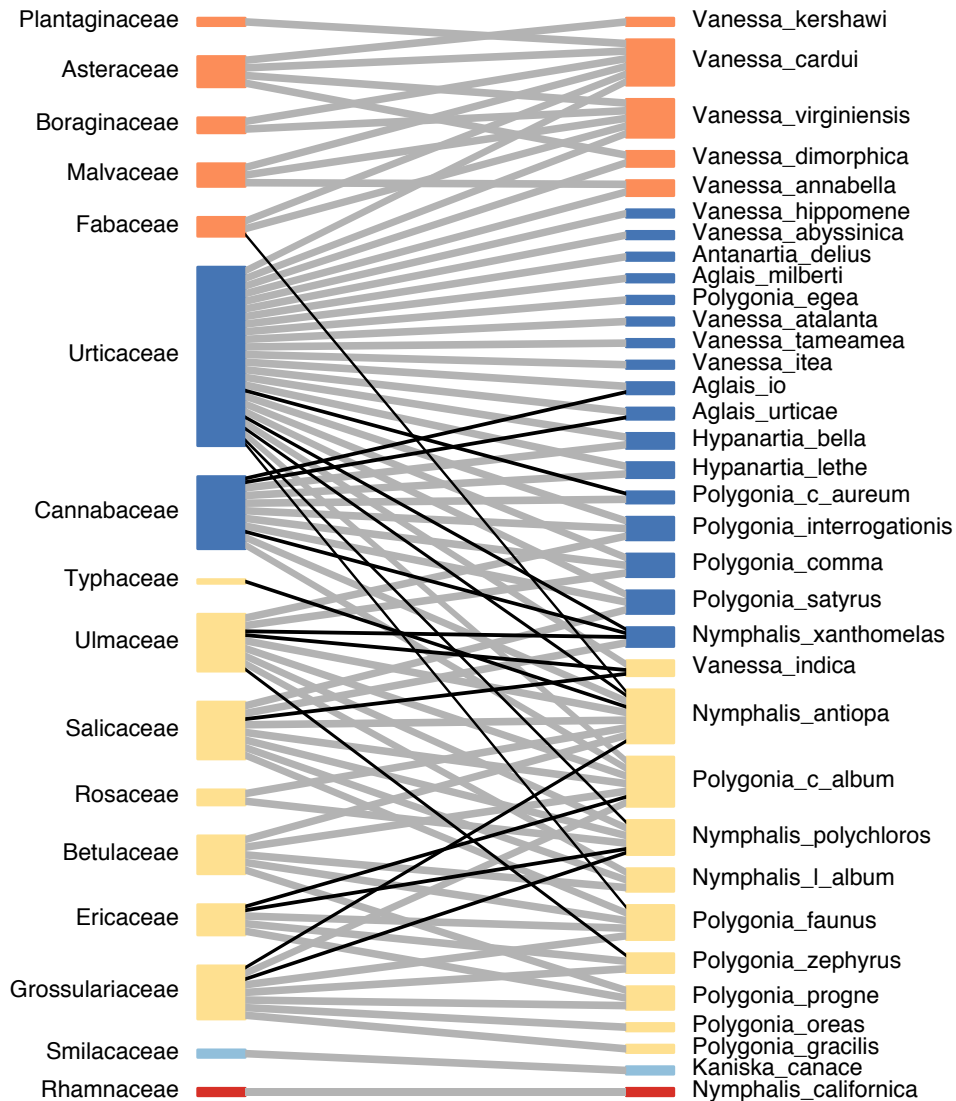


Figure 2: Interactions between Nymphalini butterflies (right) and angiosperm families (left) with taxa ordered to show the five modules identified in this dataset (colors). Grey lines represent interactions with actual hosts and black lines represent interactions with potential hosts.

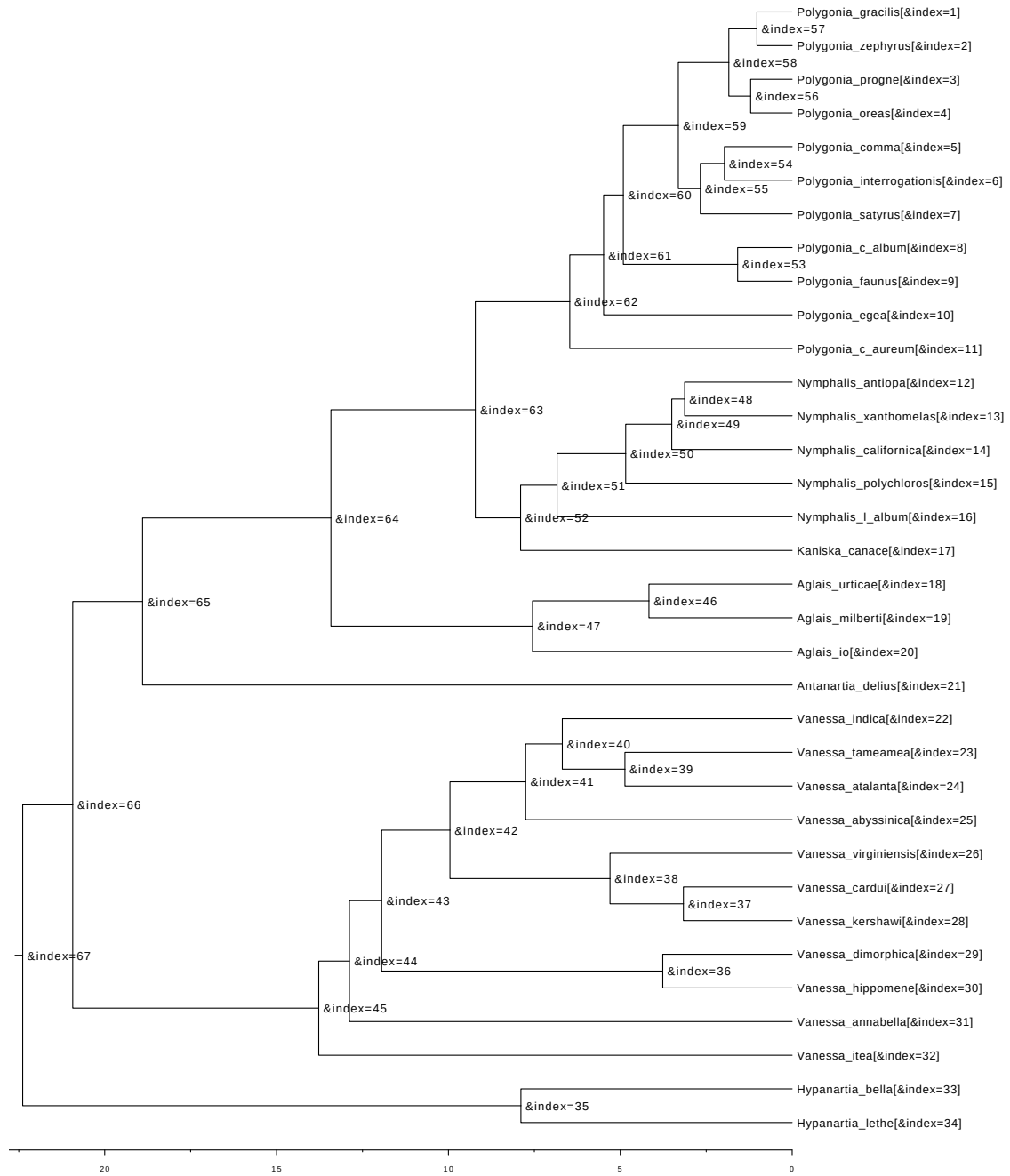


Figure 3: Time-calibrated phylogeny of Nymphalini tribe from Chazot et al. (unpublished). Indices of internal nodes correspond to row names in Figure S5.

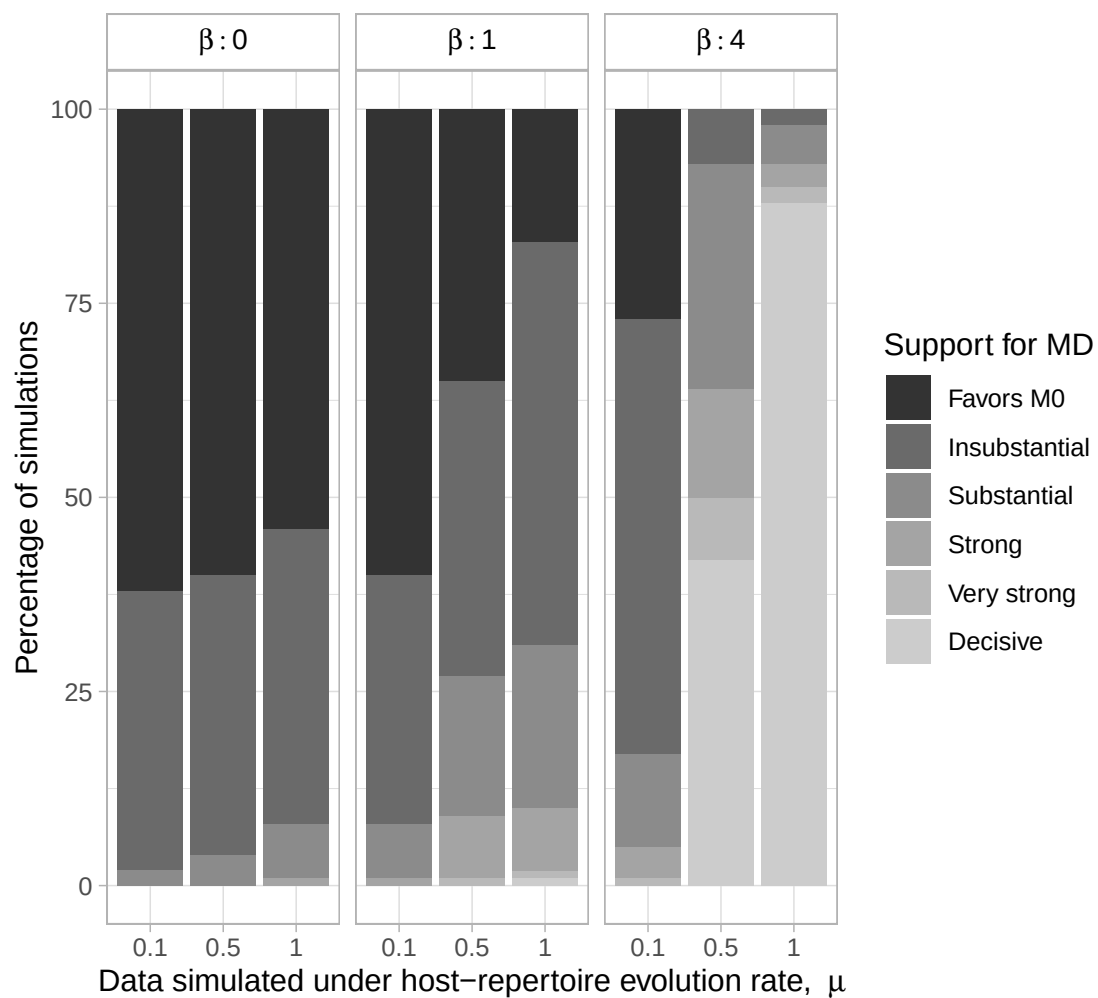


Figure 4: Distribution of Bayes factors for the simulation study when using the time-calibrated tree to calculate phylogenetic distance between host lineages. Each column corresponds to the strength of support per 2×50 MCMC analyses.

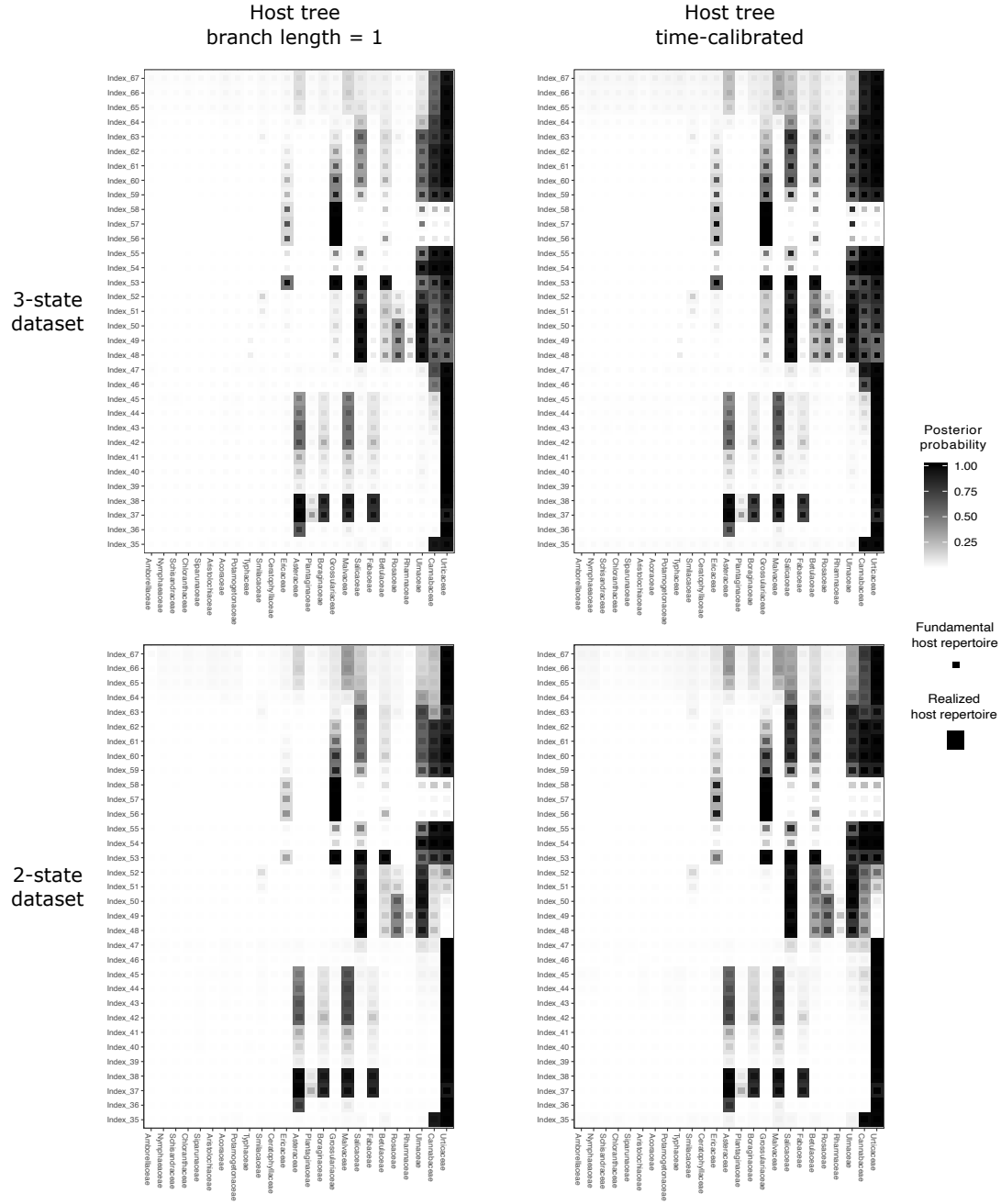


Figure 5: Posterior probability of ancestral host repertoires of Nymphalini butterflies for four different model settings (2 or 3-state dataset, host tree with branch lengths assigned to 1 or proportional to divergence time). In each panel, rows show the host repertoires at internal nodes of the Nymphalini phylogeny (same index as in Fig. S3). Small squares represent hosts in the fundamental host repertoire (states 1 and 2), whereas large squares represent hosts in the realized host repertoire (state 2). The probabilities of a host being on the fundamental and on the realized repertoire are shown by the color scale.

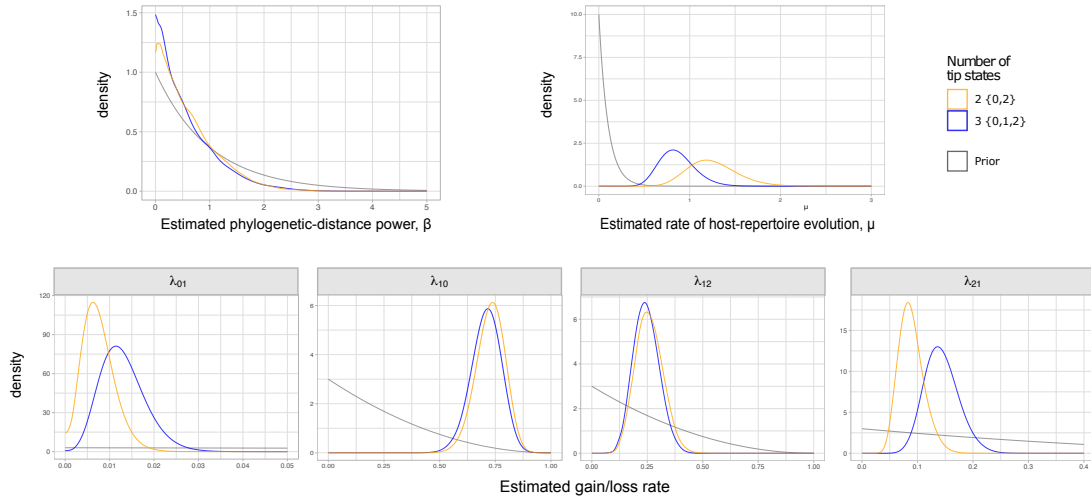


Figure 6: Marginal posterior densities for parameters in the Nymphalini-Angiosperms study when using the time-calibrated tree to calculate phylogenetic distance between host lineages. Results are shown for both the full dataset (3 states at tips) and the dataset omitting the intermediate state (2 states at tips). Grey lines corresponds to the priors $\beta \sim \text{Exponential}(1)$, $\mu \sim \text{Exponential}(10)$, and $\lambda \sim \text{Dirichlet}(1, 1, 1, 1)$.